

Raw Dataset Structure

The CONTENTS Procedure

Data Set Name	WORK.HR_RAW	Observations	54808
Member Type	DATA	Variables	13
Engine	V9	Indexes	0
Created	05/04/2026 16:51:54	Observation Length	120
Last Modified	05/04/2026 16:51:54	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	51
First Data Page	1
Max Obs per Page	1090
Obs in First Data Page	1061
Number of Data Set Repairs	0
Filename	/saswork/SAS_work92450001CEF3_odaws01-apse1.oda.sas.com/SAS_workCB580001CEF3_odaws01-apse1.oda.sas.com/hr_raw.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	2390075
Access Permission	rw-r--r--
Owner Name	u64492942
File Size	7MB
File Size (bytes)	6815744

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
8	age	Num	8	BEST12.	BEST32.
12	avg_training_score	Num	8	BEST12.	BEST32.
11	awards_won?	Num	8	BEST12.	BEST32.
2	department	Char	17	\$17.	\$17.
4	education	Char	16	\$16.	\$16.
1	employee_id	Num	8	BEST12.	BEST32.
5	gender	Char	1	\$1.	\$1.
13	is_promoted	Num	8	BEST12.	BEST32.
10	length_of_service	Num	8	BEST12.	BEST32.
7	no_of_trainings	Num	8	BEST12.	BEST32.
9	previous_year_rating	Num	8	BEST12.	BEST32.
6	recruitment_channel	Char	8	\$8.	\$8.
3	region	Char	9	\$9.	\$9.

First 10 Records — Raw Data

Obs	employee_id	department	region	education	gender	recruitment_channel	no_of_trainings	age	previous_year_rating	length_of_service	awards_won?	avg_training_score	is_promoted
1	65438	Sales & Marketing	region_7	Master's & above	f	sourcing	1	35	5	8	0	49	0
2	65141	Operations	region_22	Bachelor's	m	other	1	30	5	4	0	60	0
3	7513	Sales & Marketing	region_19	Bachelor's	m	sourcing	1	34	3	7	0	50	0
4	2542	Sales & Marketing	region_23	Bachelor's	m	other	2	39	1	10	0	50	0
5	48945	Technology	region_26	Bachelor's	m	other	1	45	3	2	0	73	0
6	58896	Analytics	region_2	Bachelor's	m	sourcing	2	31	3	7	0	85	0
7	20379	Operations	region_20	Bachelor's	f	other	1	31	3	5	0	59	0
8	16290	Operations	region_34	Master's & above	m	sourcing	1	33	3	6	0	63	0
9	73202	Analytics	region_20	Bachelor's	m	other	1	28	4	5	0	83	0
10	28911	Sales & Marketing	region_1	Master's & above	m	sourcing	1	32	5	5	0	54	0

Phase 2: Cleaned Data — Column Distributions

The FREQ Procedure

education	Frequency	Percent
Bachelor's	36669	66.90
Below Secondary	805	1.47

education	Frequency	Percent
Master's & above	14925	27.23
Unknown	2409	4.40

score_band	Frequency	Percent
Average (60-69)	11073	20.20
Excellent (80+)	10524	19.20
Good (70-79)	7343	13.40
Poor (<60)	25868	47.20

exp_band	Frequency	Percent
Junior (1-3 yrs)	18264	33.32
Mid (4-8 yrs)	25836	47.14
Senior (9+ yrs)	10708	19.54

age_group	Frequency	Percent
Experienced (	8952	16.33
Mid (30-39)	28030	51.14
Veteran (50+)	3373	6.15
Young (20-29)	14453	26.37

promo_label	Frequency	Percent
Not Prom	50140	91.48
Promoted	4668	8.52

Null Check After Cleaning — All NMISS Should Be 0

The MEANS Procedure

Variable	N Miss	N
previous_year_rating	0	54808
avg_training_score	0	54808
age	0	54808
length_of_service	0	54808
no_of_trainings	0	54808

Query 1: Overall Promotion Summary

Obs	total_employees	total_promoted	total_not_promoted	overall_promo_rate_pct
1	54808	4668	50140	8

Query 2: Department-wise Promotion Analysis

Obs	department	total_employees	promoted	promo_rate_pct	avg_training_score	avg_rating	total_award_winners
1	Procurement	7138	688	10	70	4	171
2	Operations	11348	1023	10	60	4	262
3	Analytics	5352	512	10	84	4	130
4	Technology	7138	768	10	80	4	185
5	Finance	2536	206	8	60	4	70
6	Sales & Marketing	16840	1213	8	50	4	361
7	HR	2418	136	6	50	4	45
8	R&D	999	69	6	84	4	21
9	Legal	1039	53	6	60	4	25

Query 3: Training Score Band vs Promotion Rate

Obs	score_band	total_employees	promoted	promo_rate_pct
1	Excellent (80+)	10524	1729	16
2	Good (70-79)	7343	783	10
3	Average (60-69)	11073	941	8
4	Poor (<60)	25868	1215	4

Query 4: Award Winners vs Non-Winners Promotion Rate

Obs	award_status	total_employees	promoted	promo_rate_pct
1	No Award	53538	4109	8

Obs	award_status	total_employees	promoted	promo_rate_pct
2	Award Winner	1270	559	44

Query 5: Recruitment Channel Effectiveness

Obs	recruitment_channel	total_employees	promoted	promo_rate_pct	avg_score
1	referred	1142	138	12	66
2	other	30446	2556	8	64
3	sourcing	23220	1974	8	64

Query 5: Recruitment Channel Effectiveness

total_at_risk_employees
8803

Query 6: At-Risk Employees — Requires HR Intervention

Obs	employee_id	department	region	age	avg_training_score	previous_year_rating	length_of_service	risk_reason
1	42747	Sales & Marketing	region_15	35	39	1	8	Low Score + No Award + Not Promoted
2	51473	Sales & Marketing	region_15	34	39	1	8	Low Score + No Award + Not Promoted
3	14391	Sales & Marketing	region_28	32	40	5	3	Low Score + No Award + Not Promoted
4	46166	Sales & Marketing	region_2	51	40	5	19	Low Score + No Award + Not Promoted
5	17417	HR	region_2	39	40	5	10	Low Score + No Award + Not Promoted
6	48497	Sales & Marketing	region_2	24	40	3	1	Low Score + No Award + Not Promoted
7	30185	Sales & Marketing	region_30	44	40	1	7	Low Score + No Award + Not Promoted
8	3981	Sales & Marketing	region_33	30	41	2	4	Low Score + No Award + Not Promoted
9	19924	Sales & Marketing	region_11	32	41	5	6	Low Score + No Award + Not Promoted
10	1859	Sales & Marketing	region_13	50	41	5	19	Low Score + No Award + Not Promoted
11	49491	Sales & Marketing	region_16	25	41	1	2	Low Score + No Award + Not Promoted
12	40439	Sales & Marketing	region_2	37	41	1	10	Low Score + No Award + Not Promoted
13	55946	Sales & Marketing	region_2	32	41	1	5	Low Score + No Award + Not Promoted
14	18051	Sales & Marketing	region_7	48	41	4	9	Low Score + No Award + Not Promoted
15	45931	Sales & Marketing	region_22	33	41	5	5	Low Score + No Award + Not Promoted
16	71069	Sales & Marketing	region_21	35	41	4	7	Low Score + No Award + Not Promoted
17	26470	Sales & Marketing	region_20	35	41	3	3	Low Score + No Award + Not Promoted
18	56647	Sales & Marketing	region_11	26	41	3	3	Low Score + No Award + Not Promoted
19	35711	Sales & Marketing	region_2	52	41	3	12	Low Score + No Award + Not Promoted
20	74453	Sales & Marketing	region_12	27	41	3	2	Low Score + No Award + Not Promoted

FREQ 1: Education Level vs Promotion — Chi-Square Test

The FREQ Procedure

Frequency Col Pct	Table of education by promo_label			
	education	promo_label		
		Not Prom	Promoted	Total
	Bachelor's	33661 67.13	3008 64.44	36669
	Below Secondary	738 1.47	67 1.44	805
	Master's & above	13454 26.83	1471 31.51	14925
	Unknown	2287 4.56	122 2.61	2409
	Total	50140	4668	54808

Statistics for Table of education by promo_label

Statistic	DF	Value	Prob
Chi-Square	3	75.8734	<.0001
Likelihood Ratio Chi-Square	3	80.1645	<.0001
Mantel-Haenszel Chi-Square	1	5.0597	0.0245
Phi Coefficient		0.0372	
Contingency Coefficient		0.0372	
Cramer's V		0.0372	

Sample Size = 54808

FREQ 2: Gender vs Promotion — Chi-Square Test

The FREQ Procedure

Frequency Col Pct	Table of gender by promo_label			
	promo_label			
	gender	Not Prom	Promoted	Total
	f	14845 29.61	1467 31.43	16312
	m	35295 70.39	3201 68.57	38496
	Total	50140	4668	54808

Statistics for Table of gender by promo_label

Statistic	DF	Value	Prob
Chi-Square	1	6.7640	0.0093
Likelihood Ratio Chi-Square	1	6.7005	0.0096
Continuity Adj. Chi-Square	1	6.6773	0.0098
Mantel-Haenszel Chi-Square	1	6.7639	0.0093
Phi Coefficient		-0.0111	
Contingency Coefficient		0.0111	
Cramer's V		-0.0111	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	14845
Left-sided Pr <= F	0.0050
Right-sided Pr >= F	0.9954
Table Probability (P)	0.0005
Two-sided Pr <= P	0.0100

Sample Size = 54808

FREQ 3: Experience Band vs Promotion

The FREQ Procedure

Frequency Col Pct	Table of exp_band by promo_label			
	promo_label			
	exp_band	Not Prom	Promoted	Total
	Junior (1-3 yrs)	16683 33.27	1581 33.87	18264
	Mid (4-8 yrs)	23629 47.13	2207 47.28	25836
	Senior (9+ yrs)	9828 19.60	880 18.85	10708
	Total	50140	4668	54808

FREQ 4: Age Group vs Promotion

The FREQ Procedure

Frequency Col Pct	Table of age_group by promo_label			
	promo_label			
	age_group	Not Prom	Promoted	Total
	Experienced (	8288 16.53	664 14.22	8952
	Mid (30-39)	25474 50.81	2556 54.76	28030
	Veteran (50+)	3151 6.28	222 4.76	3373
	Young (20-29)	13227 26.38	1226 26.26	14453
	Total	50140	4668	54808

FREQ 5: Training Score Band vs Promotion

The FREQ Procedure

Frequency Col Pct	Table of score_band by promo_label			
	promo_label			
	score_band	Not Prom	Promoted	Total
	Average (60-69)	10132 20.21	941 20.16	11073

Table of score_band by promo_label			
score_band	promo_label		
	Not Prom	Promoted	Total
Excellent (80+)	8795 17.54	1729 37.04	10524
Good (70-79)	6560 13.08	783 16.77	7343
Poor (<60)	24653 49.17	1215 26.03	25868
Total	50140	4668	54808

FREQ 6: Recruitment Channel vs Promotion — Chi-Square Test

The FREQ Procedure

Frequency Col Pct	Table of recruitment_channel by promo_label			
	recruitment_channel	promo_label		
		Not Prom	Promoted	Total
	other	27890 55.62	2556 54.76	30446
	referred	1004 2.00	138 2.96	1142
	sourcing	21246 42.37	1974 42.29	23220
	Total	50140	4668	54808

Statistics for Table of recruitment_channel by promo_label

Statistic	DF	Value	Prob
Chi-Square	2	19.2363	<.0001
Likelihood Ratio Chi-Square	2	17.2786	0.0002
Mantel-Haenszel Chi-Square	1	0.2722	0.6018
Phi Coefficient		0.0187	
Contingency Coefficient		0.0187	
Cramer's V		0.0187	

Sample Size = 54808

MEANS 1: Promoted vs Not Promoted — Statistical Comparison

The MEANS Procedure

promo_label	N Obs	Variable	N	Mean	Median	Std Dev	Minimum	Maximum
Not Prom	50140	age	50140	34.84	33.00	7.71	20.00	60.00
		avg_training_score	50140	62.65	60.00	12.99	39.00	96.00
		previous_year_rating	50140	3.25	3.00	1.21	1.00	5.00
		length_of_service	50140	5.88	5.00	4.29	1.00	37.00
		no_of_trainings	50140	1.26	1.00	0.62	1.00	10.00
Promoted	4668	age	4668	34.37	33.00	7.08	20.00	60.00
		avg_training_score	4668	71.33	71.00	14.75	41.00	99.00
		previous_year_rating	4668	3.91	4.00	1.05	1.00	5.00
		length_of_service	4668	5.72	5.00	4.01	1.00	34.00
		no_of_trainings	4668	1.20	1.00	0.51	1.00	6.00

MEANS 2: Department-wise Training Score & Rating Summary

The MEANS Procedure

department	N Obs	Variable	N	Mean	Median
Analytics	5352	avg_training_score	5352	84.60	85.00
		previous_year_rating	5352	3.51	3.00
Finance	2536	avg_training_score	2536	60.22	60.00
		previous_year_rating	2536	3.45	3.00
HR	2418	avg_training_score	2418	50.02	50.00
		previous_year_rating	2418	3.43	3.00
Legal	1039	avg_training_score	1039	59.87	60.00
		previous_year_rating	1039	3.34	3.00
Operations	11348	avg_training_score	11348	60.23	60.00
		previous_year_rating	11348	3.60	4.00
Procurement	7138	avg_training_score	7138	70.12	70.00
		previous_year_rating	7138	3.28	3.00
R&D	999	avg_training_score	999	84.60	85.00
		previous_year_rating	999	3.51	3.00
Sales & Marketing	16840	avg_training_score	16840	50.26	50.00
		previous_year_rating	16840	3.06	3.00
Technology	7138	avg_training_score	7138	79.93	80.00
		previous_year_rating	7138	3.15	3.00

MEANS 3: High Performer Flag vs Promotion

The MEANS Procedure

high_performer	N Obs	Variable	N	Mean	Sum
0	47802	is_promoted	47802	0.07	3196.00
		avg_training_score	47802	60.82	2907389.00
		previous_year_rating	47802	3.13	149406.00
1	7006	is_promoted	7006	0.21	1472.00
		avg_training_score	7006	80.89	566712.00
		previous_year_rating	7006	4.53	31706.00

HR Report 1: Overall Promotion Summary

Total Employees	Total Promoted	Not Promoted	Promo Rate %
54808	4668	50140	8.00

Source: HR Employee Dataset | 54,808 records

HR Report 2: Department-wise Promotion Ranking

Department	Total Emp	Promoted	Promo Rate %	Avg Score	Avg Rating	Award Winners
Procurement	7138	688	10.00	70.00	4.00	171
Operations	11348	1023	10.00	60.00	4.00	262
Analytics	5352	512	10.00	84.00	4.00	130
Technology	7138	768	10.00	80.00	4.00	185
Finance	2536	206	8.00	60.00	4.00	70
Sales & Marketing	16840	1213	8.00	50.00	4.00	361
HR	2418	136	6.00	50.00	4.00	45
R&D	999	69	6.00	84.00	4.00	21
Legal	1039	53	6.00	60.00	4.00	25

Green = High Performing | Red = Needs HR Attention

HR Report 3: Training Score Band vs Promotion Rate

Score Band	Total Emp	Promoted	Promo Rate %
Excellent (80+)	10524	1729	16.00
Good (70-79)	7343	783	10.00
Average (60-69)	11073	941	8.00
Poor (<60)	25868	1215	4.00

Employees scoring 80+ are 4x more likely to be promoted

HR Report 4: Award Winners vs Non-Winners Promotion Impact

Award Status	Total Emp	Promoted	Promo Rate %
No Award	53538	4109	8.00
Award Winner	1270	559	44.00

Award winners are 5.7x more likely to be promoted

HR Report 5: Recruitment Channel Effectiveness

Channel	Total Emp	Promoted	Promo Rate %	Avg Score
referred	1142	138	12.00	66.00
other	30446	2556	8.00	64.00
sourcing	23220	1974	8.00	64.00

Referred employees have highest promotion rate at 12.1%

HR Report 6: At-Risk Employees — Immediate HR Attention Required

Emp ID	Department	Age	Score	Rating	Service Yrs	Risk Flag
42747	Sales & Marketing	35	39	1.0	8	Low Score + No Award + Not Promoted
51473	Sales & Marketing	34	39	1.0	8	Low Score + No Award + Not Promoted
14391	Sales & Marketing	32	40	5.0	3	Low Score + No Award + Not Promoted
46166	Sales & Marketing	51	40	5.0	19	Low Score + No Award + Not Promoted
17417	HR	39	40	5.0	10	Low Score + No Award + Not Promoted
48497	Sales & Marketing	24	40	3.0	1	Low Score + No Award + Not Promoted
30185	Sales & Marketing	44	40	1.0	7	Low Score + No Award + Not Promoted

Emp ID	Department	Age	Score	Rating	Service Yrs	Risk Flag
3981	Sales & Marketing	30	41	2.0	4	Low Score + No Award + Not Promoted
19924	Sales & Marketing	32	41	5.0	6	Low Score + No Award + Not Promoted
1859	Sales & Marketing	50	41	5.0	19	Low Score + No Award + Not Promoted
49491	Sales & Marketing	25	41	1.0	2	Low Score + No Award + Not Promoted
40439	Sales & Marketing	37	41	1.0	10	Low Score + No Award + Not Promoted
55946	Sales & Marketing	32	41	1.0	5	Low Score + No Award + Not Promoted
18051	Sales & Marketing	48	41	4.0	9	Low Score + No Award + Not Promoted
45931	Sales & Marketing	33	41	5.0	5	Low Score + No Award + Not Promoted
71069	Sales & Marketing	35	41	4.0	7	Low Score + No Award + Not Promoted
26470	Sales & Marketing	35	41	3.0	3	Low Score + No Award + Not Promoted
56647	Sales & Marketing	26	41	3.0	3	Low Score + No Award + Not Promoted
35711	Sales & Marketing	52	41	3.0	12	Low Score + No Award + Not Promoted
74453	Sales & Marketing	27	41	3.0	2	Low Score + No Award + Not Promoted
74079	Sales & Marketing	25	41	3.0	1	Low Score + No Award + Not Promoted
53032	Sales & Marketing	36	41	5.0	4	Low Score + No Award + Not Promoted
66465	Sales & Marketing	27	41	3.0	1	Low Score + No Award + Not Promoted
56263	Sales & Marketing	38	41	1.0	7	Low Score + No Award + Not Promoted
74647	Sales & Marketing	31	41	4.0	3	Low Score + No Award + Not Promoted
14954	Sales & Marketing	36	41	5.0	2	Low Score + No Award + Not Promoted
54265	HR	39	41	4.0	4	Low Score + No Award + Not Promoted
36532	Sales & Marketing	30	41	3.0	2	Low Score + No Award + Not Promoted
55267	Sales & Marketing	41	41	4.0	6	Low Score + No Award + Not Promoted
25223	Sales & Marketing	27	41	2.0	4	Low Score + No Award + Not Promoted
69342	Sales & Marketing	26	41	1.0	4	Low Score + No Award + Not Promoted
30341	Sales & Marketing	32	41	3.0	2	Low Score + No Award + Not Promoted
21179	Sales & Marketing	30	42	3.0	6	Low Score + No Award + Not Promoted
61378	HR	24	42	3.0	1	Low Score + No Award + Not Promoted
21723	Sales & Marketing	37	42	1.0	6	Low Score + No Award + Not Promoted
35624	Sales & Marketing	29	42	3.0	4	Low Score + No Award + Not Promoted
44593	Sales & Marketing	33	42	5.0	7	Low Score + No Award + Not Promoted
66191	Sales & Marketing	53	42	4.0	12	Low Score + No Award + Not Promoted
63521	Sales & Marketing	44	42	5.0	3	Low Score + No Award + Not Promoted
67140	Sales & Marketing	32	42	3.0	5	Low Score + No Award + Not Promoted
9336	Sales & Marketing	31	42	3.0	7	Low Score + No Award + Not Promoted
77289	Sales & Marketing	31	42	3.0	6	Low Score + No Award + Not Promoted
36107	Sales & Marketing	36	42	3.0	10	Low Score + No Award + Not Promoted
335	Sales & Marketing	27	42	3.0	1	Low Score + No Award + Not Promoted
60511	Sales & Marketing	23	42	3.0	1	Low Score + No Award + Not Promoted
68145	Sales & Marketing	24	42	5.0	2	Low Score + No Award + Not Promoted
22369	Sales & Marketing	34	42	4.0	6	Low Score + No Award + Not Promoted
6432	Sales & Marketing	44	42	3.0	14	Low Score + No Award + Not Promoted
23706	Sales & Marketing	44	42	2.0	5	Low Score + No Award + Not Promoted
62893	Sales & Marketing	30	42	4.0	4	Low Score + No Award + Not Promoted

Top 50 lowest scoring employees with no awards — needs intervention

